



WELCOME

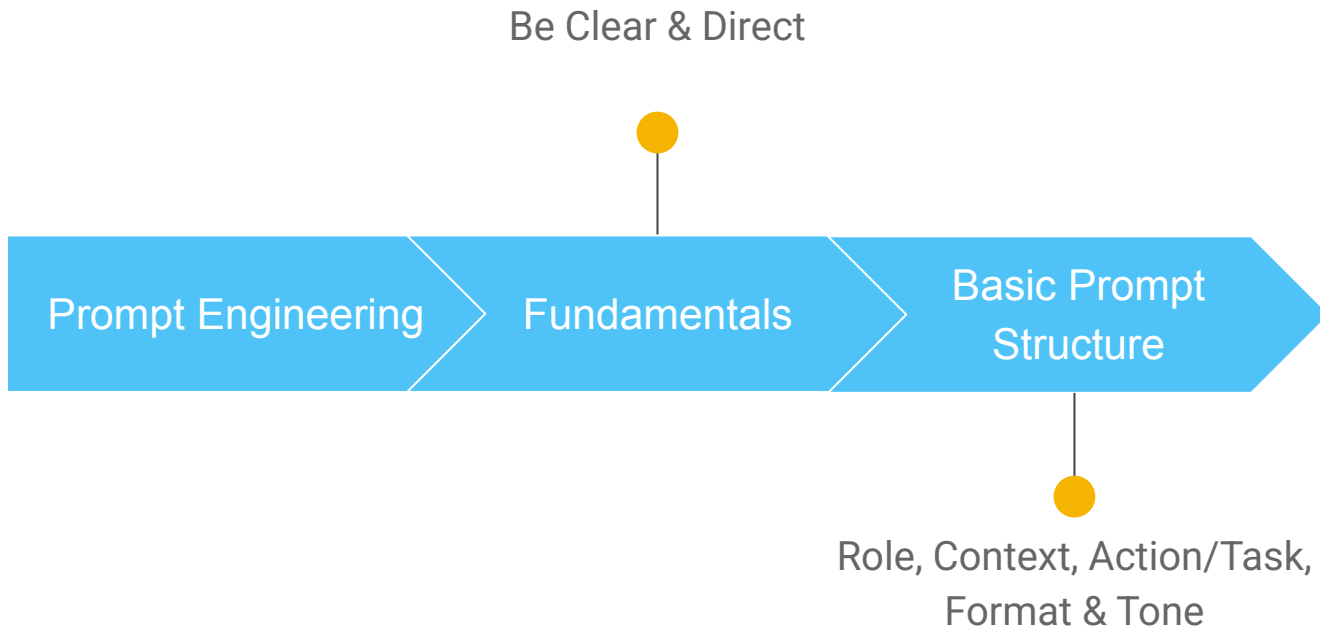




Prompt Engineering Fundamentals

Agenda for Today's Session

Prompt Engineering Fundamentals



Question

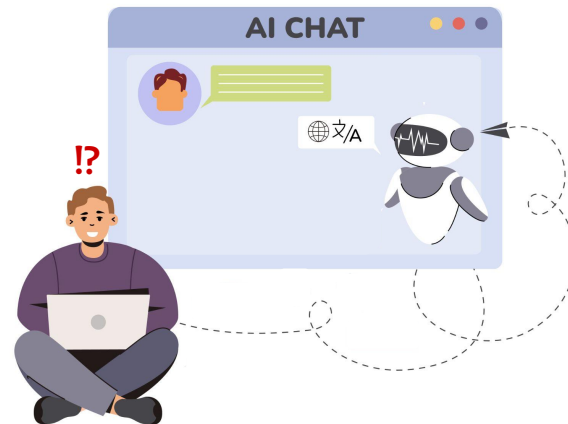


Have you ever asked an LLM
a question and received
a confusing or unhelpful response?



Better Prompt... Better Results..!

You must have noticed that the difference is likely dependent on **how you asked the question**



Generative AI

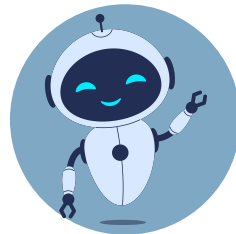
Prompt

The **natural language text**, which **describes** the **task** that an LLM should perform, is called a **prompt**



Describe the country “India” in two lines

India is a vast South Asian country known for its incredibly diverse culture, languages, and religions. It is also the world's most populous democracy, characterized by a rich history and rapid economic growth



Generative AI

Prompt - Example



What is generative AI?



Chat GPT



Gemini



LLAMA 3



Claude

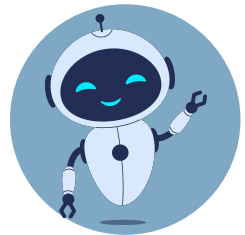
Generative AI Prompt



I'm a first-year B.Tech student.

Can you explain what generative AI is in simple terms, with examples of how it's used in apps like ChatGPT?

An explanation including what generative AI means, AI models that create new content (like text, images, or music) based on learning patterns from data, with examples



Prompt

Example 2



How to prepare for exams?



Chat GPT



Gemini



LLAMA 3



Claude

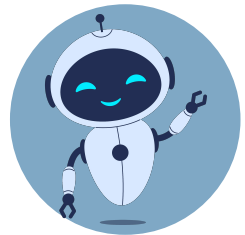
Prompt

Example 2



I'm a first-year B.Tech student with exams in two weeks. Can you recommend a study plan that includes revision strategies, breaks, and how to handle difficult subjects?

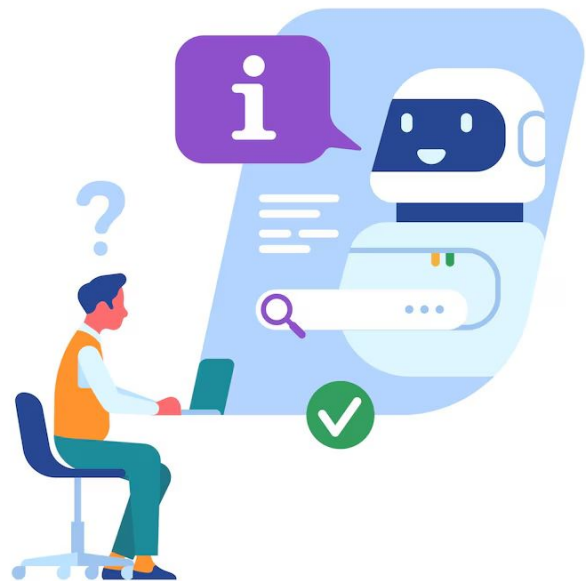
A detailed study plan suggests dividing the syllabus by days, active recall techniques, regular short breaks, focusing extra time on tough subjects, and sleep importance.



Observation

If you observe carefully, the LLM response depends on

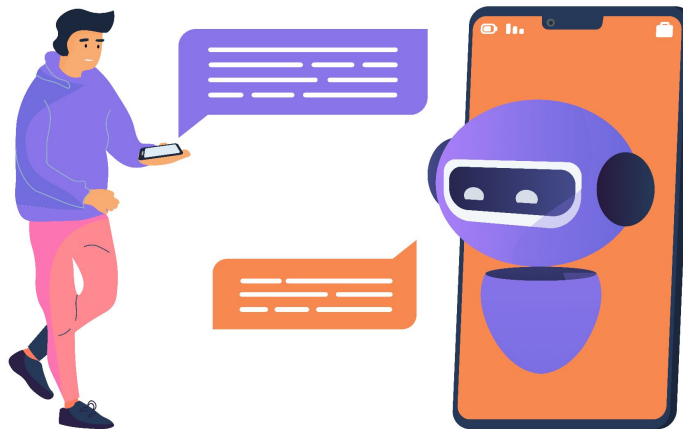
how the question is asked



Prompt

Better Prompts Lead To Better Results!

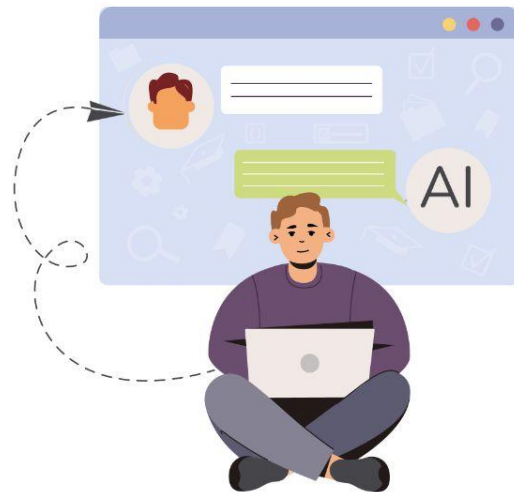
The clearer and more detailed your prompt is,
the better the model output will be



Prompt Engineering

Prompt Engineering

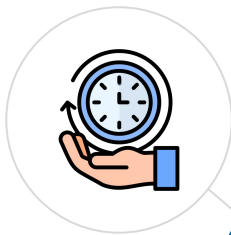
Prompt engineering is the **art of creating prompts** that help an LLM generate more accurate and higher-quality answers



Why Prompt Engineering?

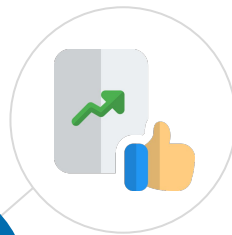
Saves time

Get what you need
in less iterations



Better results

More accurate, relevant, and
useful responses



Avoid confusion

A clear prompt prevents
the model from making
incorrect assumptions



Consistency

Well-engineered prompts
produce reliable results
each time



Prompt
Engineering

Fundamentals

Fundamentals

Be Clear and Direct



Think of an LLM like
a new Intern



Doesn't know what to do
unless you explain



Needs detailed instructions
to perform correctly

Be Clear and Direct



The more clearly and specifically you explain
what you want to the LLM,
the better and more accurate response will be

Fundamentals

Be Clear and Direct



Write a resume.



Can you try Improvising the Prompt?



Improvised Prompt



Can you help me create a professional resume for a BTech CSE student? Include sections like personal details, education, skills, internships, projects, and achievements. Please provide tips for each section to make the resume stand out to potential employers

Prompting Frameworks

Different Prompting Frameworks

RCAFT

Role, Context, Action, Format, Tone

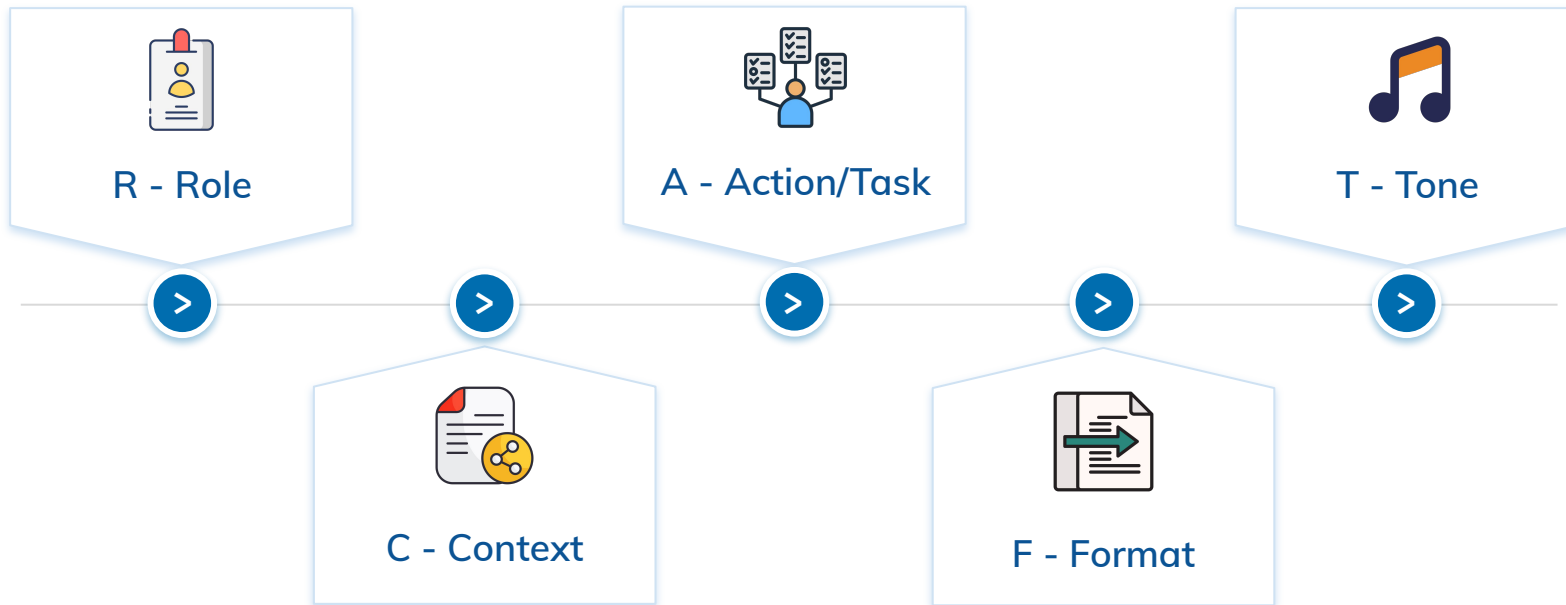
STAR

Situation, Task, Action, Result

RISEN

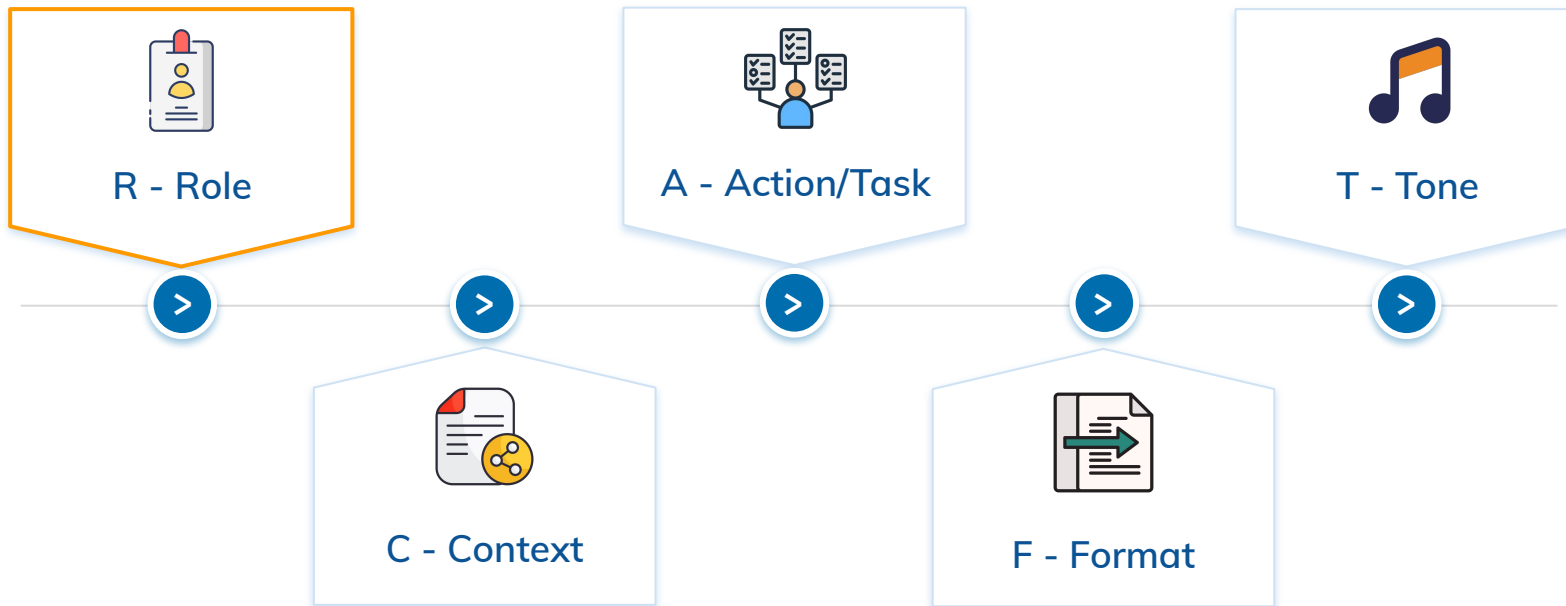
Role, Input, Steps, End Goal, Narrowing

RCAFT Framework - CRAFT (easy to remember)



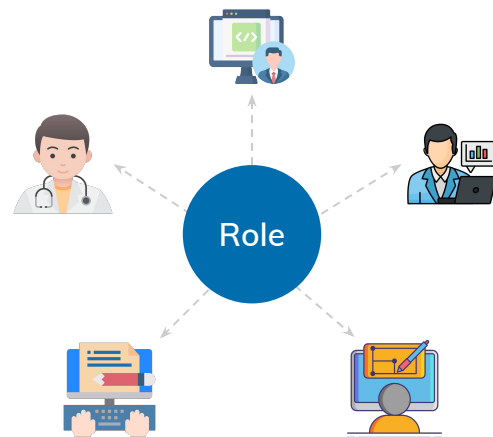
Basic Prompt Structure

Role



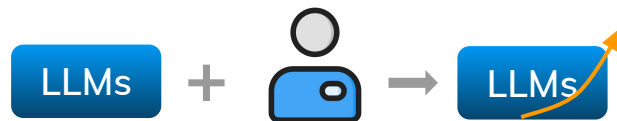
Role

- It is sometimes important to prompt the LLM to take on a specific role
- This technique is known as role prompting
- The more detail you provide about the role and context, the better the results



Role

- Priming an LLM with a specific role can enhance its performance across various tasks, from writing to coding to summarizing
- It's like how humans can sometimes be helped when told to "think like a _____"



Role

- Role prompting can also change the style, tone, and manner of the LLM's response

LLMs

style

tone

manner



Role

Examples

- 1 You are a professional chef with 20 years of experience in Italian cuisine
- 2 Act like a stand-up comedian from New York
- 3 You are a senior software engineer specializing in frontend development



**Can you write a prompt
that you would give to any LLM
to help you plan a trip?**



Role

Example



Act as an experienced travel consultant who specializes in budget-friendly trips and has personally visited over 50 destinations



Chat GPT



Gemini

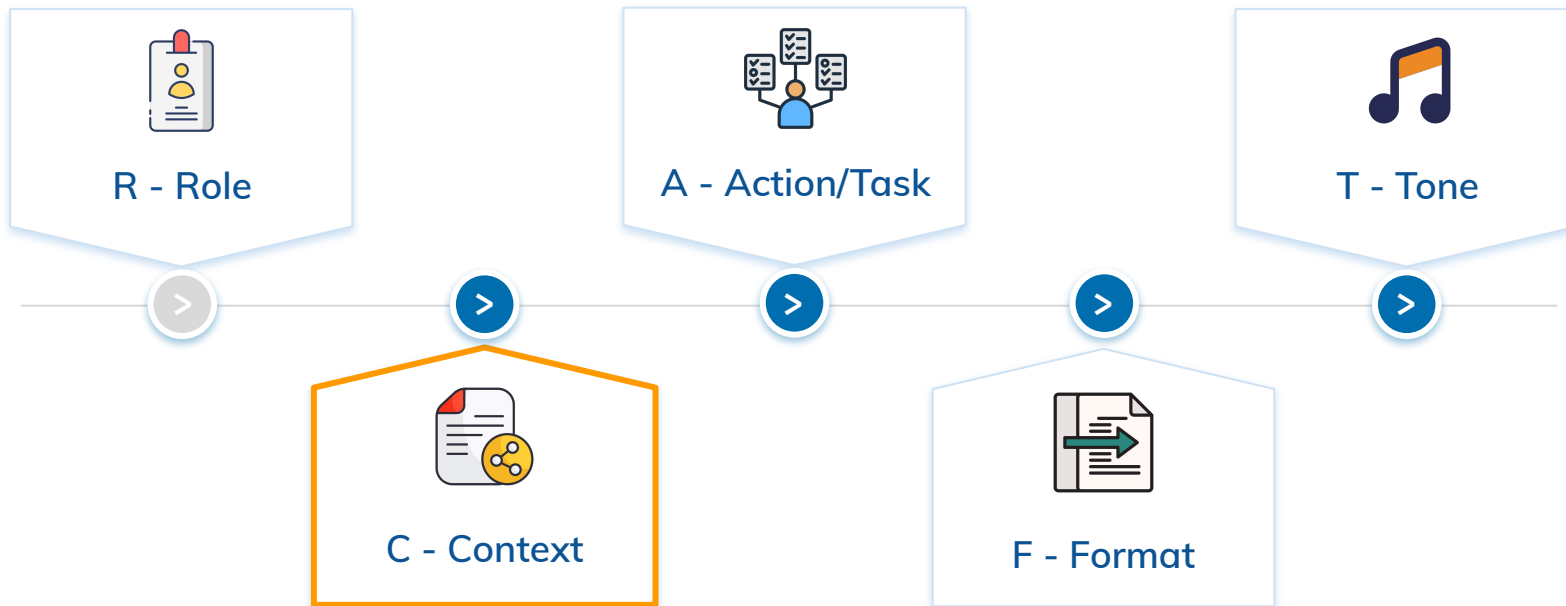


LLAMA 3



Claude

Context



Context

- Context tells the LLM **who it's helping, what the situation is, and any limitations**
- Good context helps the LLM understand your specific needs



Examples

- 1 Personal situation ("I'm planning a trip to Japan")
- 2 Audience information ("This is for high school students")
- 3 Project details ("We're launching a new product next month")
- 4 Constraints ("We have a limited budget")
- 5 Prior knowledge ("I have no experience with coding")



**Now let's provide the context
to our travel consultant**



Context

Example



Act as an experienced travel consultant who specializes in budget-friendly trips and has personally visited over 50 destinations

I'm a **22-year-old college student** planning my first solo trip. I have a **budget of 20,000, 10 days to travel in June**, and I'm interested in both **outdoor activities and experiencing local culture**. I'm **somewhat adventurous** but also **concerned about safety** as a solo traveler.

Context

Example



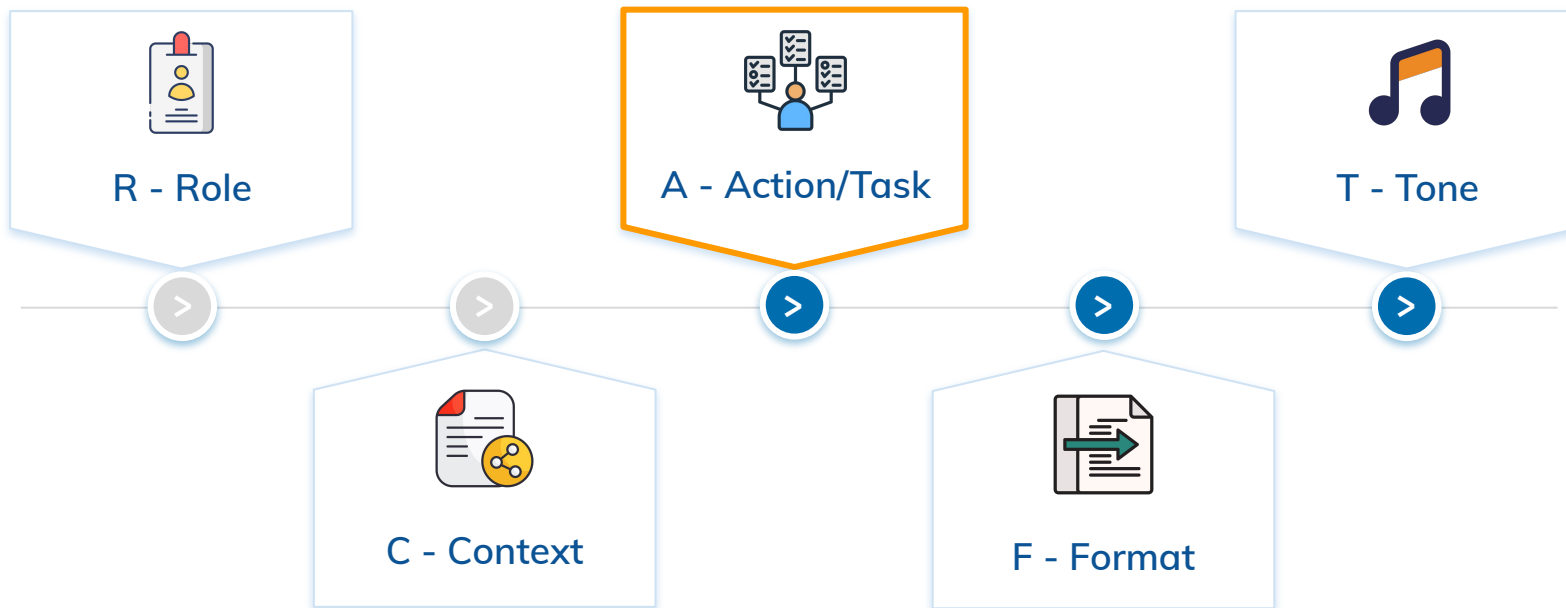
"I'm planning a surprise birthday party for my spouse who loves hiking and 80s music"



"I'm preparing for a job interview at a tech startup and need to practice answering common questions"

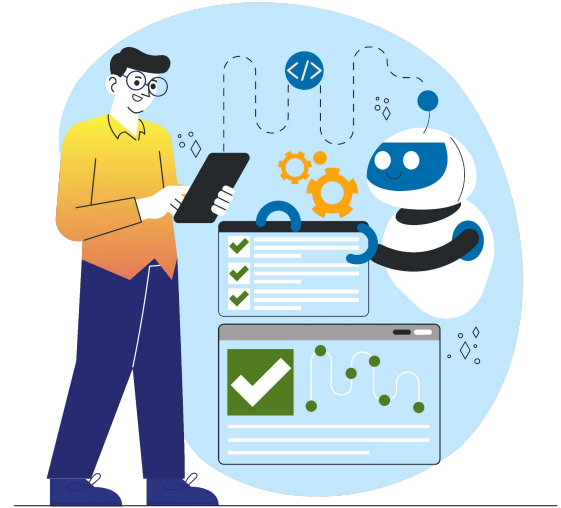
Basic Prompt Structure

Action/Task



Action/Task

- The specific action or task you want the LLM to perform
- This is the "verb" of your prompt - the specific action you want the LLM to take



Examples

- 1 Explain/teach ("Explain Generative AI")
- 2 Create/generate ("Create a marketing plan")
- 3 Analyze/evaluate ("Analyze this paragraph")
- 4 Summarize ("Summarize this research paper")
- 5 Compare/contrast ("Compare these two approaches")
- 6 Brainstorm ("Brainstorm names for my startup")

Action

Example

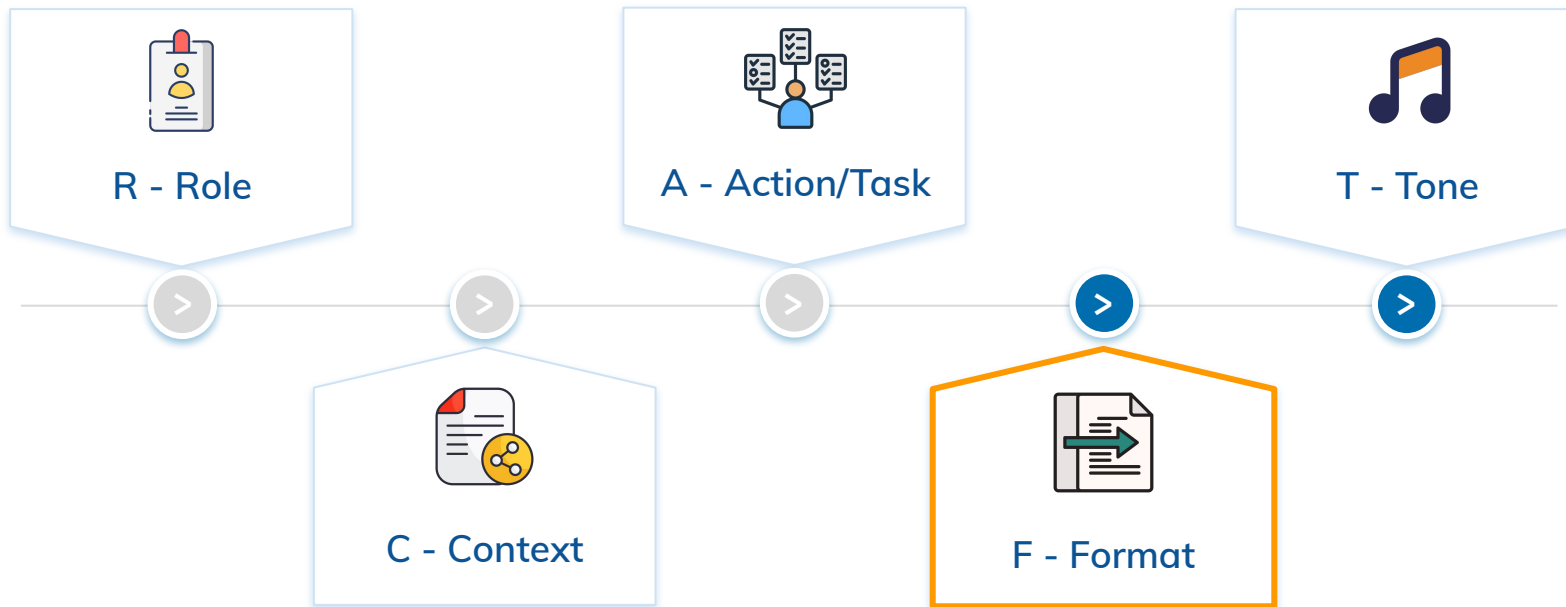


Act as an experienced travel consultant ...

I'm a 22-year-old college student planning my first solo trip...

Recommend 2 specific destinations that would be ideal for my situation, and for each destination, suggest a 10-day itinerary that balances cultural experiences, outdoor activities, and relaxation.

Format



Format

- Format tells the LLM how you want your answer organized and presented



Format

Examples

- 1 Bullet points or numbered lists
- 2 Table or chart
- 3 Step-by-step guide
- 4 FAQ style (question and answer)
- 5 Timeline
- 6 Markdown formatting with headers and subheaders

Format

Example



Act as an experienced travel consultant ...

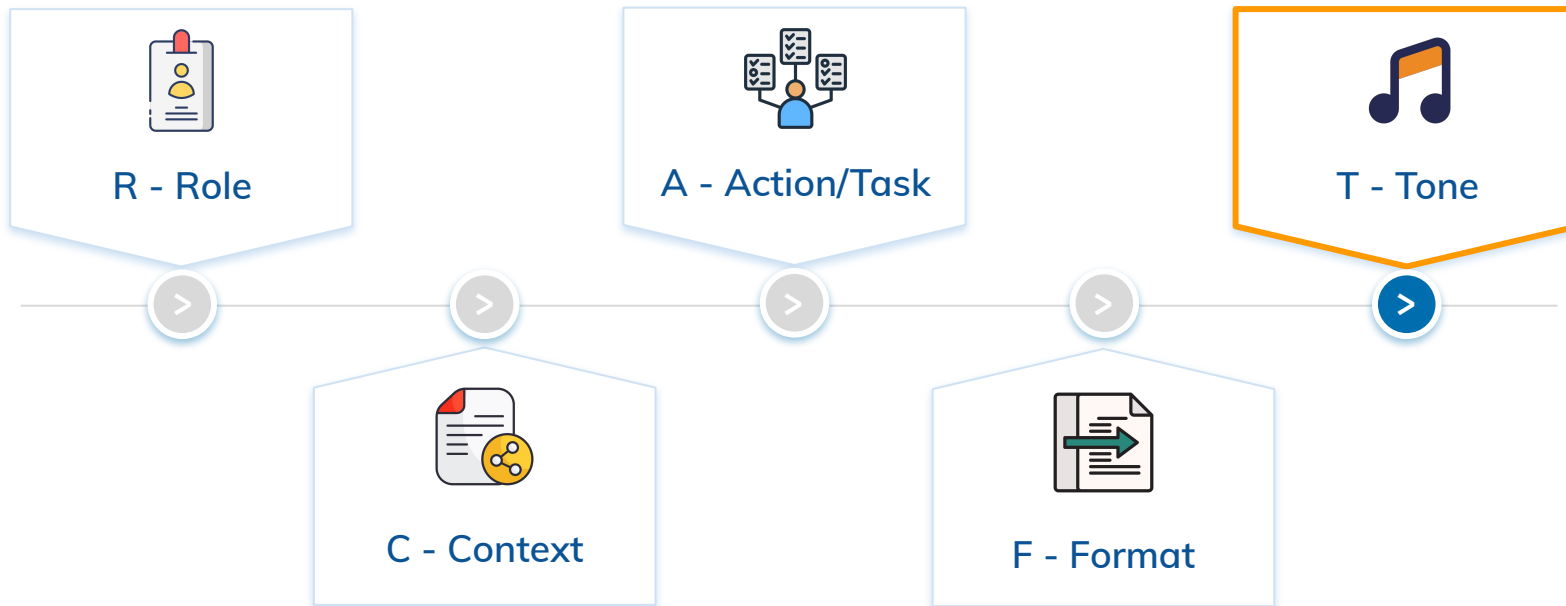
I'm a 22-year-old college student ...

Recommend 2 specific destinations ...

Format:

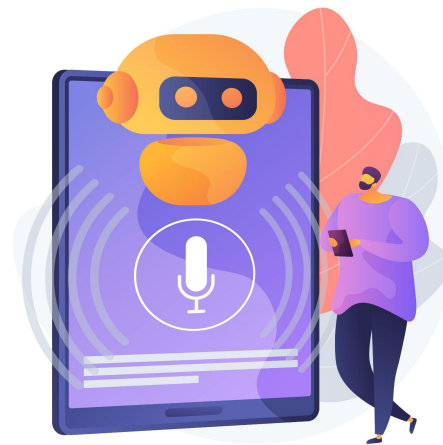
1. For each destination, create a section that includes:
 - A brief overview (3-5 sentences)
 - An Estimated cost breakdown (accommodation, food, activities, transport)
 - Safety considerations for solo travelers
2. Then provide a day-by-day itinerary in a table format with the following columns for:
 - Day #
 - Morning activity
 - Afternoon activity
 - Evening activity
 - Accommodation
3. End with a pros/cons list for each destination

Tone



Tone

- The **style, voice, or emotional quality** you want the LLM to use in its response
- This affects how the message feels to the reader



Tone

Examples

1

Formal or academic

2

Friendly and conversational

3

Encouraging and supportive

4

Direct and concise

5

Enthusiastic and energetic

6

Cautious and balanced

7

Simple and easy to understand

Tone

Example



Act as an experienced travel consultant ...

I'm a 22-year-old college student ...

Recommend 2 specific destinations ...

Format:

For each destination, create a section with:

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Morning activity ...

End with a pros/cons list for each destination

Include some humor where appropriate, and write as if you're a slightly older friend who wants to make sure I have an amazing but safe experience. Use casual language but don't skimp on important details.

Example: Personal Study Assistant



Role; You are an experienced academic advisor who has helped hundreds of students choose their career paths and specializations.

Context: I'm a first-year BTech CSE student trying to decide which technical domain to specialize in. I'm interested in AI/ML, web development, cybersecurity, and data science, but I'm confused about career prospects, required skills, and market demand for each field.

Action: Provide a detailed comparison of these four technical domains to help me make an informed decision about my specialization

Format:

Create a comparison table with the following columns:

- Domain Name
- Required Core Skills
- Average Starting Salary (India)
- Job Market Demand (High/Medium/Low)
- Major Companies Hiring
- Typical Career Progression (5-year outlook)
- Learning Difficulty (1-10 scale)
- Future Growth Potential

Prompting is an Iterative Process



**AS YOU
NOTICE**

The best way to prompt engineering
is to **iterate**

By changing persona, changing parameters,
adding relevant examples

Prompting Techniques



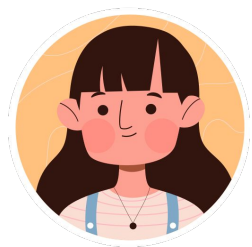
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When you got it to a point where it looks good,
create a template

So that, next time we don't need to go through
these iterations

Example Prompt Templates

Interview Preparation



Template: "You're an experienced career counselor. I have an upcoming {{INTERVIEW_TYPE}} interview for a {{JOB_ROLE}} position. Provide me with 8-10 targeted practice questions I should prepare for, along with strategic tips for answering them effectively. Present this as a clear study guide with practical examples I can practice."

Inputs:

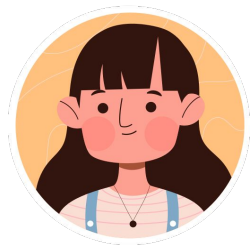
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Final Prompt:

"You're an experienced career counselor helping job seekers succeed. I have an upcoming **technical** interview for a **software engineer** position. Provide me with 8-10 targeted practice questions I should prepare for, along with strategic tips for answering them effectively. Present this as a clear study guide with practical examples I can practice."

Example Prompt Templates

Summarizing an Article



Template: "You're a research assistant skilled at distilling complex information. I need to quickly understand the key insights from `{{ARTICLE_TOPIC}}`. Read through the content and create a concise summary that captures the main arguments, supporting evidence, and conclusions. Format this as bullet points with the most important takeaways first, using clear and objective language."

Inputs:

- `{{ARTICLE_TOPIC}}`: "climate change research paper" / "new technology trends report" / "healthcare policy analysis" / "market research study" / "historical analysis piece"

Final Prompt:

"You're a research assistant skilled at distilling complex information. I need to quickly understand the key insights from **climate change research paper**. Read through the content and create a concise summary that captures the main arguments, supporting evidence, and conclusions. Format this as bullet points, placing the most important takeaways first, using clear and objective language."

Quiz Time!

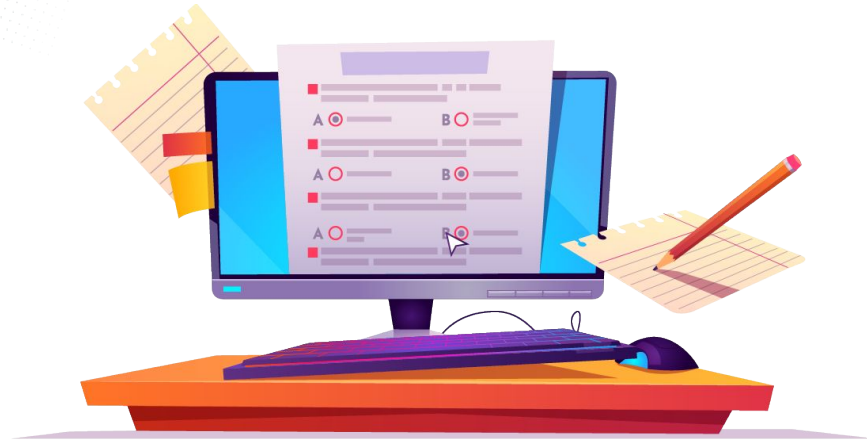


Key Takeaways

- ✓ Prompt Engineering
- ✓ Prompt Structure
 - Role
 - Context
 - Action/Task
 - Format
 - Tone
- ✓ Prompt Template



Practice





THANK YOU





ALL THE BEST



Format

- Format tells the LLM how you want your answer organized and presented



Format

Examples

- 1 Bullet points or numbered lists
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**Let's update the prompt such that
our travel consultant will provide a
day wise plan in a table format**



Format

Example



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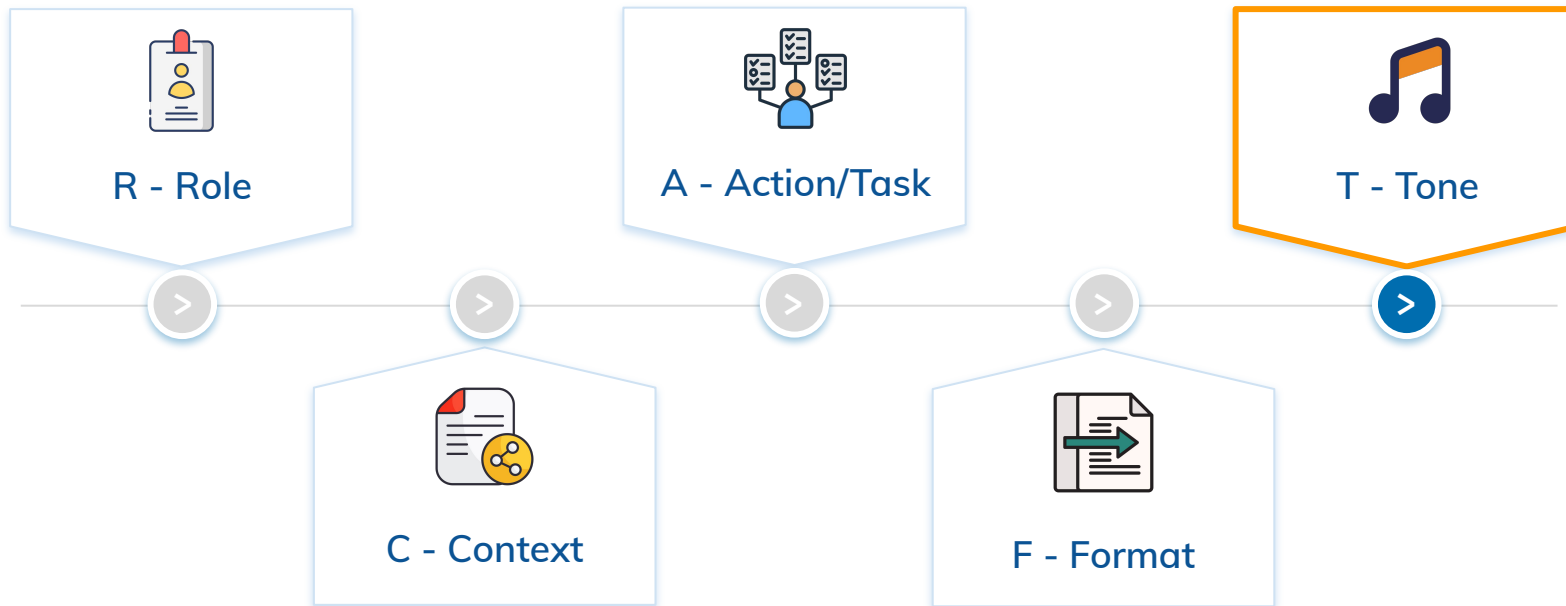
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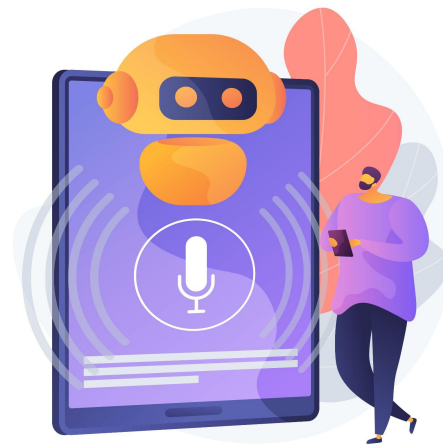
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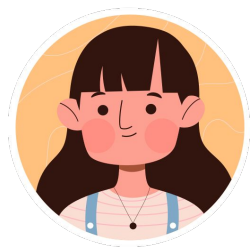
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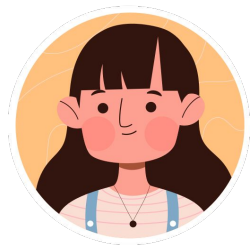
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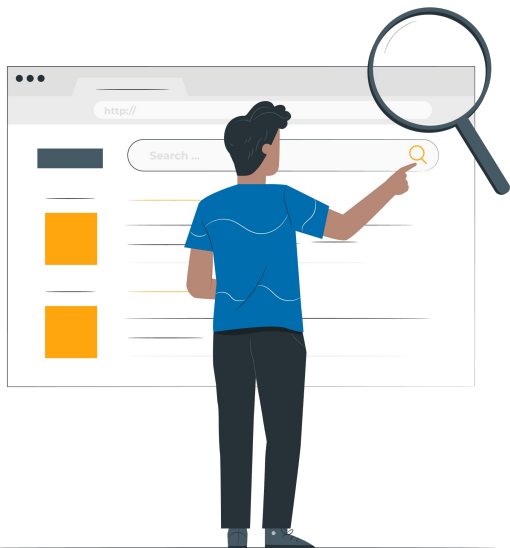
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Additional Exploration



Open AI

- <https://platform.openai.com/docs/examples>
- <https://cookbook.openai.com/>

For Cross-Platform Exploration

- <https://cloud.google.com/vertex-ai/generative-ai/docs/prompt-gallery>
- <https://github.com/microsoft/prompts-for-edu/tree/main/Students/Prompts>

Prompting Techniques

Prompting Techniques

Prompting engineering techniques allow us to **perform more complex tasks** and **enhance the reliability** and **performance** of LLMs

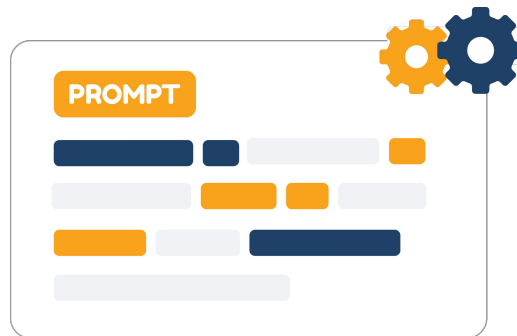


Prompting Techniques

Some well-known prompting techniques are

- ☐ Zero-shot
- ☐ One-shot
- ☐ Few-shot

Many more ...

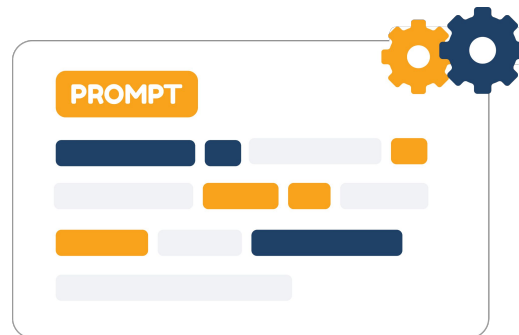


Zero-shot

Some well-known prompting techniques are

- ☐ Zero-shot 
- ☐ One-shot
- ☐ Few-shot

Many more ...



Zero-shot

Zero-shot prompting is a technique that instructs an LLM to perform a task **without providing any examples**



Zero-shot

Example



Summarize the following paragraph about Generative AI:

Generative AI refers to artificial intelligence systems capable of creating new content, such as text, images, music, and videos that resemble human-created work. Unlike traditional AI, which focuses on analysis and prediction, generative AI can produce original outputs based on patterns learned from training data. Popular examples include GPT models for text, DALL-E for images, and MusicLM for music creation. These systems have sparked discussions about creativity, copyright, and the future of human-machine collaboration in creative fields."

Zero-shot

Example



Classify this review as positive or negative:

The restaurant was incredible, best meal I've had all year.

Response: Positive

One-shot

Some well-known prompting techniques are

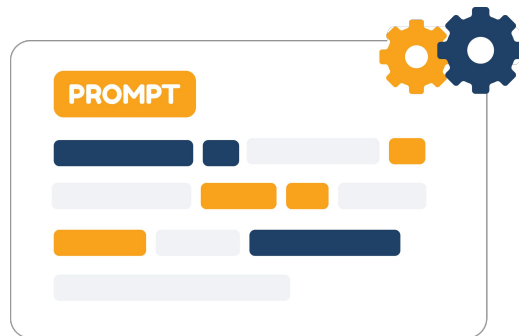
☒ Zero-shot

☐ One-shot



☐ Few-shot

Many more ...



One-shot

Providing a single example to the model so it has something to imitate in order to complete the task more effectively



One-shot

Example



"Write a birthday message for my best friend who loves bikes"

Example:

'Happy birthday, bro! May your love for bikes grow even faster than your age.'

Now write a similar message for my other friend who loves Marvel movies

One-shot

Example



Review: Food was cold and service slow.

Sentiment: Negative

Review: The desserts were heavenly.

Sentiment:

Response: Positive

Few-shot

Some well-known prompting techniques are

☒ Zero-shot

☒ One-shot

☐ Few-shot



Many more ...



Few-shot

Providing multiple examples
to the model



Few-shot

It is similar to one-shot, but we provide **multiple examples of the desired pattern** which increases the chance that the model follows the pattern



Few shot

Example



A is for Apple, a fruit that's sweet
B is for Banana, the best lunch treat

Few-shot

Example



Classify by sentiment (positive/negative/neutral):

Review: Food was cold.

Sentiment: Negative

Review: 'Great atmosphere!

Sentiment: Positive

Review: Restaurant was busy.

Sentiment: Neutral

Review: The prices were reasonable.

Sentiment:

Examples

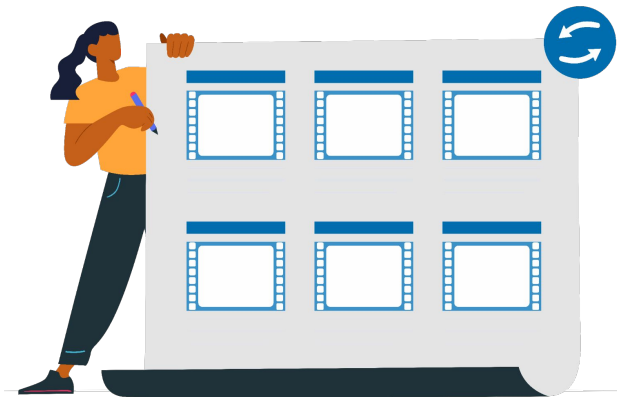
Prompting Techniques	When it is best
Zero-shot prompting	• The task is straightforward • The AI likely has seen similar examples during training • You need a quick response
One-shot prompting	• You need a specific format • The pattern is simple but needs an example
Few-shot prompting	• The task involves a more complex pattern • You need consistent formatting across multiple outputs • The task requires specialized or domain-specific knowledge

Tricks and Tips

Tricks and Tips

Separating Data from Instructions

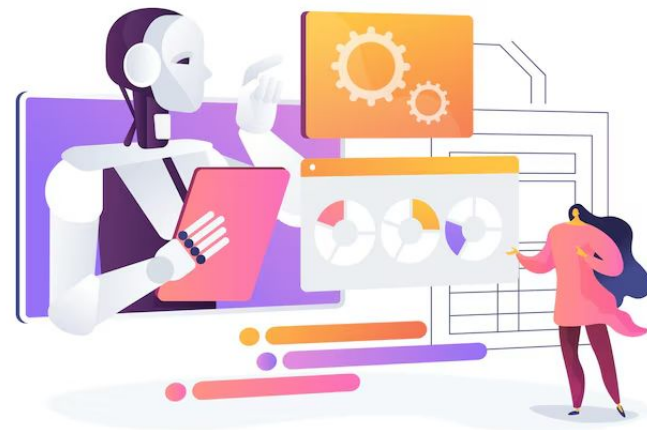
Instead of writing a new prompt each time, you can create a reusable prompt template and that can be reused with different data each time



Tricks and Tips

Separating Data from Instructions

This is especially useful when you need the LLM to perform the same task repeatedly but with different input data



Example 1: Simple Template



I will tell you the name of an animal. Please respond with
the sound that animal makes. {{ANIMAL}}

Why Separate Data from Instructions



Reusability

Create once, use multiple times with different data



Consistency

Same task structure ensures consistent results



Clarity

Keeps instructions separate from variable content

Example



Template: "Translate the following text to Spanish: {{TEXT}}"

Input: "Hello, how are you today?"

Input: "The weather is beautiful today"

Input: "I love learning new languages"

Final Prompt:

"Translate the following text to Spanish: **Hello, how are you today?**"

Example



Template: "Explain {{CONCEPT}} at a {{LEVEL}} level with examples."

Inputs:

- {{CONCEPT}}: "recursion in programming" / "object-oriented programming" / "Ohm's Law"
- {{LEVEL}}: "high school" / "first year btech student" / "intermediate" / "advanced"

Final Prompt:

"Explain **recursion in programming** at a **high school** level with examples."

Quiz Time!



Interview Preparation



Template: "You're an experienced career counselor helping job seekers succeed. I have an upcoming {{INTERVIEW_TYPE}} interview for a {{JOB_ROLE}} position. Provide me with 8-10 targeted practice questions I should prepare for, along with strategic tips for answering them effectively. Present this as a clear study guide with practical examples I can practice."

Inputs:

- {{INTERVIEW_TYPE}}: "technical" / "behavioral" / "panel" / "phone screening" / "final round"
- {{JOB_ROLE}}: "software engineer" / "marketing manager" / "data analyst" / "sales representative" / "project manager"

Final Prompt: "You're an experienced career counselor helping job seekers succeed. I have an upcoming technical interview for a software engineer position. Provide me with 8-10 targeted practice questions I should prepare for, along with strategic tips for answering them effectively. Present this as a clear study guide with practical examples I can practice."

Summarizing an Article



Template: "You're a research assistant skilled at distilling complex information. I need to quickly understand the key insights from {{ARTICLE_TOPIC}}. Read through the content and create a concise summary that captures the main arguments, supporting evidence, and conclusions. Format this as bullet points with the most important takeaways first, using clear and objective language."

Inputs:

- {{ARTICLE_TOPIC}}: "climate change research paper" / "new technology trends report" / "healthcare policy analysis" / "market research study" / "historical analysis piece"

Final Prompt: "You're a research assistant skilled at distilling complex information. I need to quickly understand the key insights from climate change research paper. Read through the content and create a concise summary that captures the main arguments, supporting evidence, and conclusions. Format this as bullet points, placing the most important takeaways first, using clear and objective language."

Example



Template: "Translate the following text to Spanish: {{TEXT}}"

Input: "Hello, how are you today?"

Input: "The weather is beautiful today"

Input: "I love learning new languages"

Final Prompt: "Translate the following text to Spanish: Hello, how are you today?"

Feedback





Next Session

Introduction to AI workflows

Structuring Prompt

System Prompt

A system prompt is a way to **provide context, instructions, and guidelines to the LLM** before you ask a question



System Prompt



"Your response should always be a series of critical thinking questions that advance the conversation"

Do not directly answer the user's question"

System Prompt

Why is the sky blue?

Prompt

Prompt Engineering

Think of it like this

- ✓ If an LLM is like a **highly knowledgeable** intern on their first day
- ✓ Prompt engineering is the skill of giving that intern **clear, detailed instructions**



Prompt

Example 2



I'm a first-year B.Tech student with exams in two weeks. Can you recommend a study plan that includes revision strategies, breaks, and how to handle difficult subjects?

A detailed study plan suggests dividing the syllabus by days, active recall techniques, regular short breaks, focusing extra time on tough subjects, and sleep importance.

